

Release Notes

for S32K311_SBAF_HSEB_RTM_0.15.0

Rev. 1.0 — 31 August 2023

Release notes
COMPANY CONFIDENTIAL

1 Getting Started

1.1 Package content

This package contains the NXP S32K311 SBAF 0.15.0:

- FULL_MEM SBAF: encrypted binary
 - s32k3x1_Secure_Baf_0.12.0_0.15.0.6_pb230804.bin.pink
- The license.txt file and the `uninstall.exe` utility for removing the SBAF binaries
- SBAF_S32K311_0_0_15_0_ReleaseNotes.pdf - this file



2 Release Details

This is the HSE Secure-BAF (SBAF) 0.15.0 RTM release for S32K311 for FULL_MEM configuration. The SBAF release of AB_SWAP configuration is part of the HSE firmware AB_SWAP pink image. Both releases can be used for production.

It is strongly recommended that user should update the SBAF in their existing samples if the features provided in this release are important to them.

The SBAF can be updated only in FULL_MEM configuration. In case of AB_SWAP, the SBAF gets updated as part of the HSE firmware update.

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP:

- S32K311
- S32K310

2.2 Version Number Details

Version number is of 8 bytes size and can be read at HSE GPR address: 0x4039C020 - - 0x4039C027

FULL_MEM:

00 0C 00 00 00 0F 00 06

AB_SWAP:

01 0C 00 00 00 0F 00 05

3 Secure-BAF Update

Secure-BAF (SBAF) can be updated via `HSE_SRV_ID_SBAF_UPDATE` and service structure `"hseSbaFUpdateSrv_t"` provided by HSE FW in one shot or in streaming mode.

The new encrypted SBAF image can be programmed at any address in the application memory before calling the SBAF update service.

The encrypted SBAF image is authenticated and decrypted, then stored in the HSE code flash memory.

On the next reset, the new SBAF boots the HSE Firmware.

Note:

- *NXP recommends updating SBAF in one shot mode.*
- *SBAF update must be performed in a stable environment because any failure in the update process causes the bricking of device.*
- *SBAF update service is only applicable for FULL_MEM configuration.*

Please refer to the **HSE Firmware Reference Manual** for more details.

4 Changes in RTM 0.15.0

Features Added

- Error-Correcting Code (ECC) handling feature: This feature will prevent the bricking of the device in case reset happens during secure flash area programming.
- Added the feature to make the desired area of application code flash area as OTP. This feature can be used only if corresponding service is provided in HSE Firmware. Currently, this feature is *not supported* by the standard HSE Firmware.
- Added the support of enabling the Firmware feature flag in Boot configuration word (Bit number 9) in addition to UTEST location 0x1B000000. The same bit can enable the firmware installation mode via MU without the need to write the marker value 0xA5 in DCM register. For more details, refer to the **HSE Firmware reference manual**.
- Added the feature in SBAF to support the firmware installation and update in backup feature enable and disable configuration via a bit 10 in Boot configuration word in IVT. This feature is applicable only for FULL_MEM configuration as in case of AB_SWAP, SBAF and HSE firmware are combined as single image. The backup disable feature is currently *not supported* in standard HSE firmware.
- Added the feature to disable the option to put application core in recovery mode in case of 8 functional and 8 destructive resets via bit number 8 in Boot configuration word in IVT. This feature is applicable only for FULL_MEM configuration as backup. In case of AB_SWAP, backup is always taken in passive area not in data flash.

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